

Effects of Dark Chocolate and Oats Sticks on Anxiety, Blood Pressure, and Heart Rate among Adolescents: A Quasi-Experimental Study

Background: Anxiety in adolescents can trigger physiological responses, including increased blood pressure and heart rate due to activation of the sympathetic nervous system. Functional food-based non-pharmacological interventions, such as dark chocolate and oats, contain bioactive compounds that may help reduce anxiety and improve cardiovascular responses. However, empirical evidence regarding the combined effects of dark chocolate and oats sticks among adolescents with anxiety remains limited. This study aimed to analyze the effect of administering dark chocolate and oats sticks on anxiety levels, blood pressure, and heart rate among adolescents with anxiety.

Methods: This study employed a quasi-experimental design with a pretest–posttest approach and a control group. A total of 60 adolescents were divided into an intervention group and a control group. Anxiety levels were assessed using a standardized questionnaire, while blood pressure and heart rate were measured using standard instruments before and after the intervention. Data were analyzed using statistical tests with a significance level of $\alpha = 0.05$.

Results: Anxiety levels decreased significantly in the intervention group compared to the control group ($p < 0.001$). In addition, systolic and diastolic blood pressure, as well as heart rate, showed significant reductions following the administration of dark chocolate and oats sticks ($p < 0.001$).

Conclusions: The administration of dark chocolate and oats sticks significantly reduced anxiety levels and improved cardiovascular physiological responses among adolescents. This intervention may serve as a practical and accessible non-pharmacological approach in promotive and preventive adolescent health programs.

Keywords: dark chocolate; oats; anxiety; blood pressure; heart rate; adolescents

INTRODUCTION

Adolescence is a developmental phase characterized by significant biological, psychological, and social changes (1). During this period, individuals are often exposed to increased academic demands, interpersonal challenges, and social expectations, which make adolescents particularly vulnerable to mental health problems, especially anxiety (2). Anxiety disorders not only affect psychological well-being but also trigger physiological responses, including increased blood pressure and heart rate, as a result of sympathetic nervous system activation (3).

The prevalence of anxiety disorders among adolescents in Indonesia remains a significant public health concern. Previous data indicate that approximately 68.7% of the Indonesian population experiences varying levels of anxiety, ranging from mild to severe (4). In addition, findings from the Indonesia National Adolescent Mental Health Survey (I-NAMHS) reported

that 3.7% of adolescents aged 10–17 years experienced anxiety disorders within a 12-month period, making it one of the most common mental health conditions in this age group (5).

Physiologically, chronic anxiety is associated with sustained activation of the sympathetic nervous system, leading to increased secretion of stress hormones such as cortisol and catecholamines. This condition results in elevated heart rate and vasoconstriction, which ultimately contributes to increased blood pressure (6). Previous studies have demonstrated a significant relationship between anxiety levels and blood pressure among adolescents, indicating that individuals with higher anxiety levels tend to exhibit higher blood pressure (7). Furthermore, epidemiological data from Indonesia show that the prevalence of prehypertension among adolescents reaches 16.8%, while hypertension accounts for 2.6% of cases, highlighting the early onset of cardiovascular risk in this population (8). Globally, the prevalence of hypertension among children and adolescents has also increased over the past two decades, rising from 3.2% to more than 6.2% (9).

Non-pharmacological approaches, particularly those based on nutritional interventions, have gained increasing attention in adolescent health research. Certain functional foods are known to influence both physiological and psychological responses. Dark chocolate, which is rich in flavonoids such as epicatechin, has been shown to enhance endothelial function through increased nitric oxide production, resulting in vasodilation and reduced blood pressure (10). Moreover, flavonoids have been associated with improved mood and reduced stress responses.

Oats, on the other hand, are a rich source of soluble fiber (beta-glucan), magnesium, and B-complex vitamins, all of which play essential roles in metabolic regulation and neurotransmitter synthesis. These components contribute to improved glucose stability and increased serotonin production, which is closely related to mood regulation and stress reduction (11). Additionally, regular oat consumption has been linked to improved cardiovascular health and reduced risk of metabolic disorders.

The combination of dark chocolate and oats presents a promising functional food intervention that may exert synergistic effects on both psychological and physiological parameters. This combination is expected to improve endothelial function, modulate autonomic nervous system activity, and enhance stress regulation (12). However, studies investigating the combined effects of these functional foods on anxiety, blood pressure, and heart rate in adolescents remain limited (13). Therefore, this study was conducted to address this gap and to provide evidence-based support for the use of simple nutritional interventions in adolescent health promotion and preventive strategies.

METHODS

This study employed a quasi-experimental design with pretest and posttest measurements. Group I received dark chocolate and oats sticks as the intervention, while Group II served as a control group and did not receive the intervention. Each group consisted of 30 participants. The study was conducted at SMA Negeri 2 Kendari in September 2021.

The study population consisted of 1,510 adolescents, with 60 respondents selected based on inclusion and exclusion criteria. Anxiety levels were measured using a questionnaire, while blood pressure and heart rate were measured using standard instruments.

Bivariate analysis was conducted using the t-test. Ethical approval was obtained (No. 80/STIKES-NH/KEPK/II/2021).

RESULT

Table 1. Characteristics of Respondents

Variable	Group I		Group II		<i>p-value</i>
	n	%	n	%	
Age (Years)					0.632
15	11	36.7	11	36.7	
16	13	43.3	14	46.7	
17	6	20.0	5	16.6	

Table 2. Anxiety Levels of Respondents Before Intervention

Variable	Group I		Group II		<i>p-value</i>
	n	%	n	%	
Anxiety					0.712
Mild	21	70.0	21	70.0	
Moderate	9	30.0	9	30.0	
Severe	0	0	0	0	
Very Severe	0	0	0	0	

Table 3. Anxiety Levels of Respondents After Intervention

Variable	Group I		Group II		<i>p-value</i>
	n	%	n	%	
Anxiety					$p < 0.001$
None	30	0	0	0	

Mild	0	0	21	70.0
Moderate	0	0	9	30.0

Table 4. Effect of Dark Chocolate Sticks on Blood Pressure in Anxious Adolescents

Group		n	Blood Pressure						<i>p-value</i>
			Baseline			Post-intervention			
			Mean	Min	Max	Mean	Min	Max	
I	Systolic	30	100	90	120	115	110	120	p < 0.001
	Diastolic	30	85	70	90	75	73	80	p < 0.001
II	Systolic	30	100	70	120	100	90	120	p < 0.001
	Diastolic	30	86	71	90	86	70	90	p < 0.001

Table 5. Effect of Dark Chocolate and Oats Sticks on Heart Rate in Anxious Adolescents

Group	n	Heart Rate						<i>p-value</i>
		Baseline			Post-intervention			
		Mean	Min	Max	Mean	Min	Max	
I	30	75	60	110	80	75	90	p < 0.001
II	30	76	60	105	70	60	102	p < 0.001

After the intervention, all respondents in Group I were categorized as not anxious, while most respondents in Group II remained mildly anxious.

Blood pressure decreased significantly in the intervention group compared to the control group ($p < 0.001$).

Heart rate showed significant changes after the intervention, indicating improved autonomic nervous system regulation ($p < 0.001$).

These findings indicate that the intervention effectively improved both psychological and physiological outcomes among adolescents.

DISCUSSION

The present study demonstrated that the administration of dark chocolate and oats sticks significantly reduced anxiety levels among adolescents. After the intervention, all respondents in the intervention group were categorized as not anxious, whereas the majority of respondents in the control group remained in the mild anxiety category. This finding indicates that the intervention had a substantial effect on improving psychological well-being.

The reduction in anxiety levels observed in this study can be explained by the bioactive compounds present in dark chocolate and oats. Dark chocolate contains flavonoids, particularly epicatechin, which have been shown to enhance cerebral blood flow and modulate neurotransmitter activity involved in mood regulation (17). Previous studies have reported that flavonoid-rich foods are associated with reduced stress responses and improved emotional states.

Oats also contribute to anxiety reduction through their nutritional composition, particularly beta-glucan and magnesium. Magnesium plays a role in inhibiting excessive neuronal excitation and has been reported to exert mild anxiolytic effects, while beta-glucan supports metabolic stability and stress response regulation (18). These mechanisms collectively contribute to improved emotional balance among adolescents.

In addition to psychological outcomes, this study found a significant reduction in systolic and diastolic blood pressure following the intervention. The decrease in blood pressure was more pronounced in the intervention group compared to the control group. These findings are consistent with previous studies demonstrating that flavonoids enhance endothelial function by increasing nitric oxide production, leading to vasodilation and reduced vascular resistance (19).

Furthermore, oats contribute to cardiovascular improvement through their beta-glucan content, which has been associated with reduced blood pressure and improved lipid metabolism. Several studies have shown that regular consumption of oats is linked to reductions in both systolic and diastolic blood pressure, particularly among individuals with elevated cardiovascular risk (20).

This study also demonstrated a significant change in heart rate following the intervention. The observed improvement in heart rate reflects better autonomic nervous system regulation, characterized by reduced sympathetic activity and increased parasympathetic dominance. This physiological response is closely associated with reduced anxiety and improved relaxation (21).

The combination of dark chocolate and oats in the form of snack sticks provides a practical and acceptable intervention for adolescents. This form not only enhances palatability but also improves adherence, which is a critical factor in the success of lifestyle-based interventions (22). The synergistic effect of these functional foods may offer a comprehensive approach to improving both mental and cardiovascular health among adolescents.

Despite these promising findings, several limitations should be acknowledged. The duration of the intervention was relatively short, and potential confounding factors such as physical activity, dietary intake, and sleep patterns were not fully controlled. Future studies are

recommended to include longer intervention periods and additional biomarkers, such as cortisol levels, to better understand the underlying mechanisms of the observed effects.

CONCLUSION

The administration of dark chocolate and oats sticks significantly reduced anxiety levels, blood pressure, and heart rate among adolescents. This functional food-based intervention may serve as a practical and non-pharmacological approach to support adolescent mental and cardiovascular health, particularly in school-based health promotion programs.

LIMITATIONS

This study did not fully control potential confounding variables such as physical activity, sleep quality, caffeine intake, and dietary patterns, which may have influenced blood pressure and heart rate outcomes.

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AUTHOR CONTRIBUTIONS

A.A., K., S.S., S., and A.M. designed research; A.A., S.S., and S. conducted research; S.S. analyzed data; A.A. and K. wrote the paper; A.A. had primary responsibility for final content. All authors read and approved the final manuscript.

CONFLICT OF INTEREST STATEMENT

The authors declare that there is no conflict of interest.

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